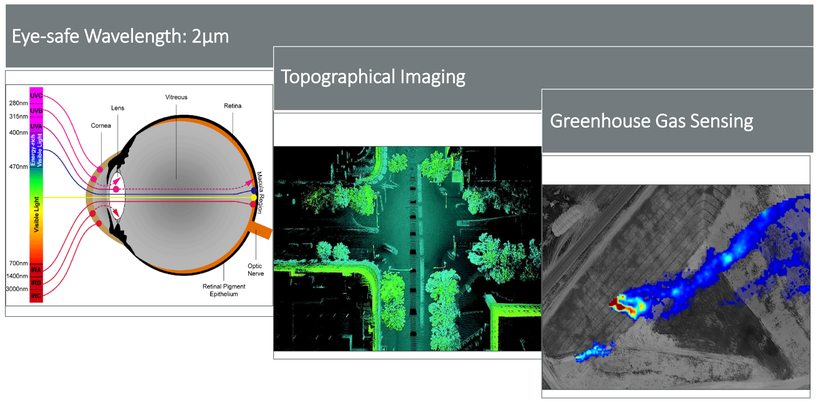
Demonstration of Zn-Diffusion in 2-µm InGaAs/GaAsSb Superlattice Planar Diodes using Atomic Layer Deposition of ZnO

M. Muduli, M. Schwartz, N. Gajowski, N. Nooman, H. Jung, S. Lee, B. Bharadwaj, and S. Krishna*

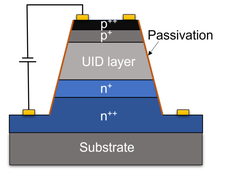
THE OHIO STATE UNIVERSITY
COLLEGE OF ENGINEERING

MOTIVATION



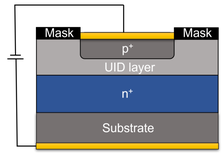
GOAL

Simplify Fabrication Process for Commercialization → Reduce Dark Current of the Detector → Preserve material quality



Generic Mesa Structure APD

(Mesa: academically preferred structure)



Generic Planar Structure APD

(Planar: commercially preferred structure)

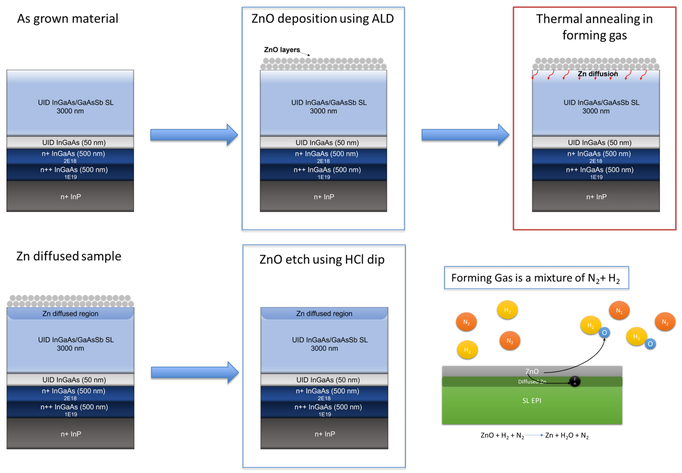
Current Approaches

Background: Approaches for dopant incorporation in planar diodes

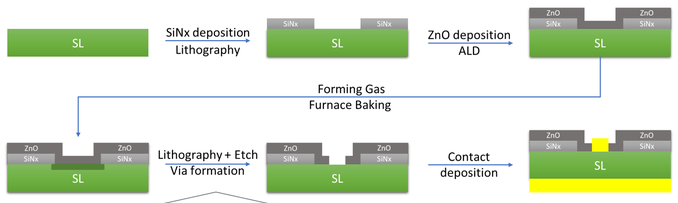
MOCVD	Ion Implantation	Thermal Diffusion	Proposed Method: ALD deposited ZnO
Pros: • Diffusion & doping is precise and accurate • Maintains material quality Cons: • Expensive • Time-consuming	Pros: • Diffusion & doping is precise and accurate Cons: • Damages the material • Expensive • Time-consuming	Pros: • Maintains material quality Cons: • Complex process	Pros: • Maintains material quality • Simple & fast process • **Novel** Cons: • Not researched enough

METHODOLOGY

DIFFUSION PROCESS

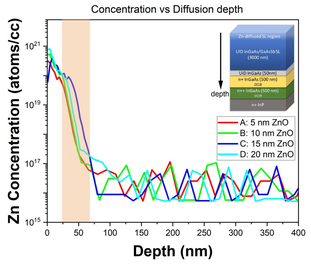


FABRICATION PROCESS



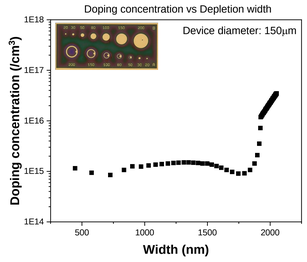
CHARACTERIZATION

SIMS



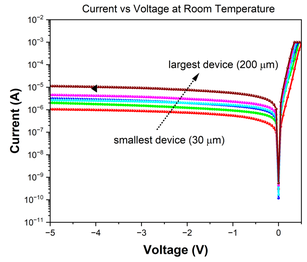
Takeaway: Zn diffusion is successful

CV Analysis



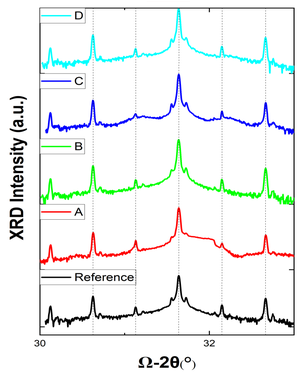
Takeaway: Background doping is ~10^{15}/cm^3

IV Measurement



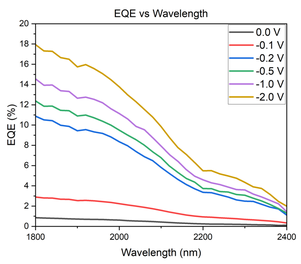
Takeaway: 1) Rectifying behavior shows Zn acts as p-dopant
2) Current is comparable to state-of-the-art***

XRD



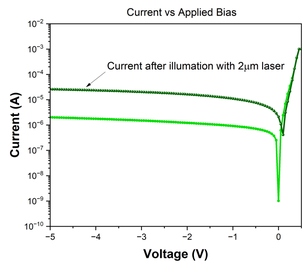
Takeaway: Material Quality Maintained

EQE Analysis



Takeaway: EQE at -1V for 2µm is 11.11%

Photocurrent Measurement



Takeaway: Device responds to 2µm wavelength

CONCLUSION

- **Achieved Zn diffusion** through the proposed method
- **Characterized the material and devices** to show material quality is maintained but the dark current is comparable to state-of-the-art***

Future Work:

- To design and demonstrate Zn-diffused APD with separate absorption charge and multiplication layer (SACM structure)
- To test devices with a guard ring structure

REFERENCES

1. Muduli, Manisha, et al. "Investigation of Zn-diffusion in 2-micron InGaAs/GaAsSb superlattice planar diodes using atomic layer deposition of ZnO." *Infrared Technology and Applications XLIX*. Vol. 12534. SPIE, 2023.
2. Martinsons, Christophe, and Georges Zissis. "Potential health issues of solid-state lighting." *IEA 4E final report* 24 (2014).
3. https://www.bridgerphotonics.com/gas-mapping-lidar-for-aerial-methane-detection
4. Uliel, Y., et al. "InGaAs/GaAsSb Type-II superlattice based photodiodes for short wave infrared detection." *Infrared Physics & Technology* 84 (2017): 63-71.
5. Ariyawansa, Gamini, et al. "Surface passivation for PhotoDetector applications." U.S. Patent No. 10,297,708. 21 May 2019.

KIND
Krishna INfrared Detector Laboratory

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George R Smith Professor of Engineering
Electrical and Computer Engineering

**state-of-the-art for Zn diffusion in InGaAs/GaAsSb

Abbreviations:-
- CV: Capacitance-Voltage
- IV: Current Voltage
- EQE: External Quantum Efficiency
- SIMS: Secondary Ion Mass Spectroscopy
- MOCVD: Metal Organic Chemical Vapor Deposition
- XRD: X-Ray Diffraction
- SL: Superlattice
- ALD: Atomic Layer Deposition
- APD: Avalanche Photodiode

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